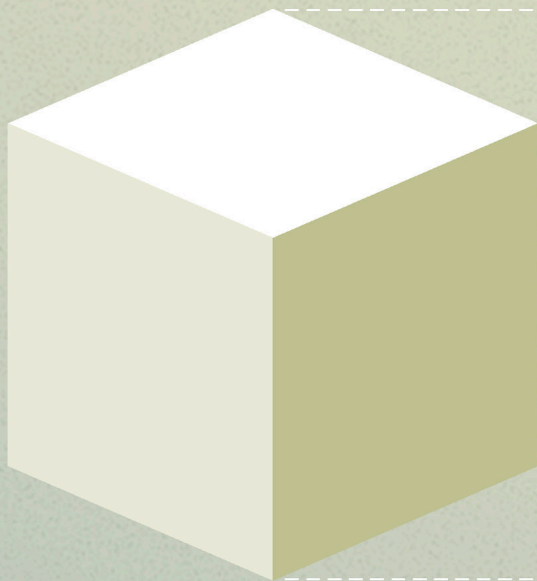


INSIDE SOLID OBJECTS

Individual atoms consist of positively charged nuclei surrounded by negatively charged electrons (see pp.10–11). In order to form large-scale solid materials, such as crystals, however, atoms surrender some of their electrons, allowing them to float freely (to a certain degree) within a fixed geometric lattice of positively charged ions. Quantum effects govern where electrons can reside—modeling the electrons as a “gas” of fermions provides insights into phenomena such as electrical and heat conduction and insulation.



SOLID OBJECT